

CSE211 Digital Design

Akdeniz University

Week12: Sequential Logic Part 1

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Course program

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Week 01	15/09/2025	Introduction
Week 02	22/09/2025	Digital Systems and Binary Numbers I
Week 03	29/09/2025	Digital Systems and Binary Numbers II
Week 04	06/10/2025	Boolean Algebra and Logic Gates I
Week 05	13/10/2025	Boolean Algebra and Logic Gates II
Week 06	20/10/2025	Gate Level Minimization
Week 07	27/10/2025	Karnaugh Maps
Week 08	27/10/2025	Karnaugh Maps
Week 09	05/11/2025	Midterm (Expected)
Week 10	17/11/2025	Combinational Logic
Week 11	24/11/2025	Combinational Logic
Week 12	01/12/2025	Timing, delays and hazards
Week 13	08/12/2025	Synchronous Sequential Logic
Week 14	15/12/2025	Synchronous Sequential Logic

Synchronous Sequential Logic

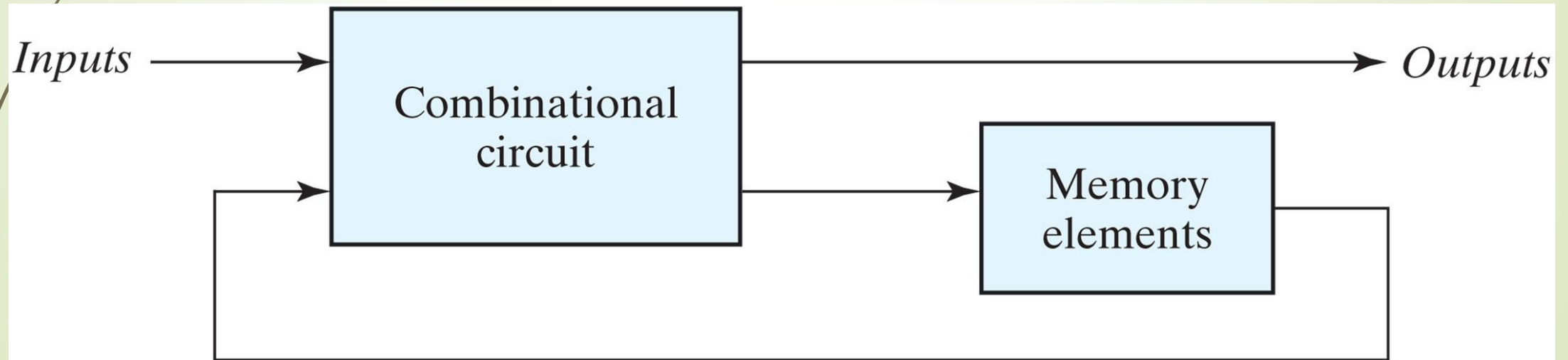
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- In combinational logic the output depends only and immediately on their inputs.
- A combinational logic cannot have some sort of **feedback** loop where the output of a logic is also the input of it.
- A **sequential network** on the other hand can have this feature.
- All consumer products, cell phones, navigation receivers, personal computers, digital cameras have the ability to send, receive, store, retrieve, and process information represented in a **binary format**.
- This is only possible with a special electronic component which can store information, **the memory**..

Block Diagram of a Sequential Circuit

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- The binary information stored in **memory elements** at any given time defines the **state** of the sequential circuit at that time.
- External inputs also determine the condition for changing the state in the storage elements
- **A sequential circuit is specified by a time sequence of inputs, outputs, and internal states**



Types of Sequential Circuits

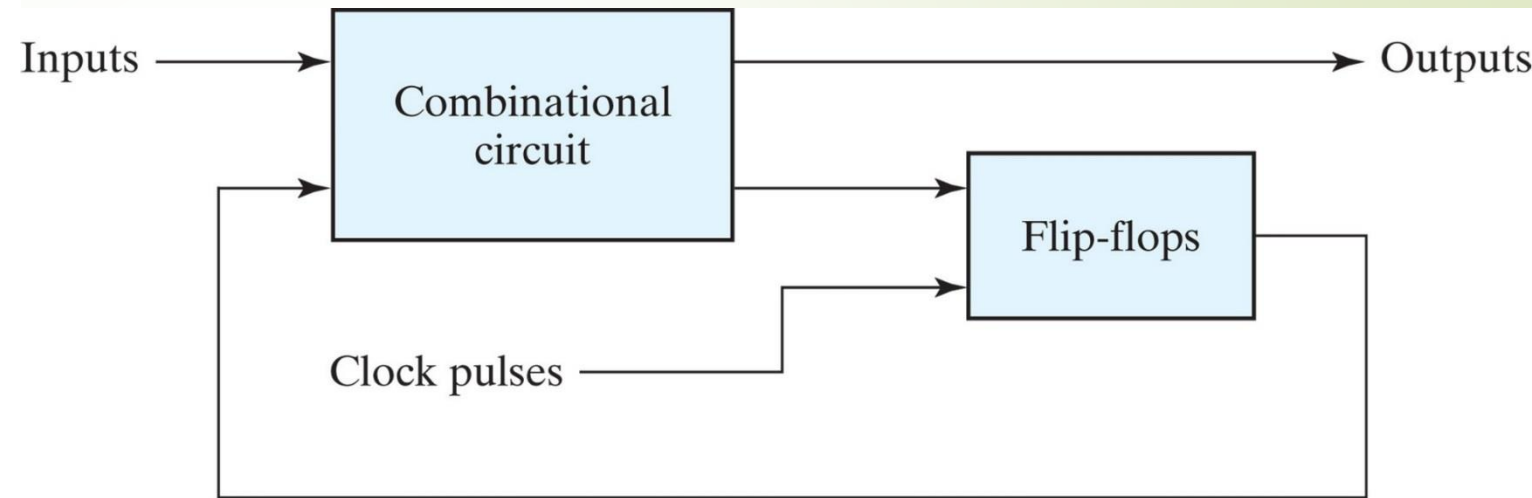
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- **Synchronous** sequential circuit
 - whose behavior can be defined from the knowledge of its signals at discrete instants of time.
 - Synchronization is achieved by a **timing device** called a **clock generator**
 - denoted by the identifiers **clock** and **clk**
 - Synchronous sequential circuits that use clock pulses to control storage elements are called **clocked sequential circuits**
 - **Less instable than asynchronous systems.**
- **Asynchronous** sequential circuit
 - behavior of an *asynchronous* sequential circuit depends upon **the input signals** at any instant of time *and* the **order** in which the inputs change
 - The storage elements commonly used in asynchronous sequential circuits are time-delay devices
 - the **internal propagation delay of logic gates** is of **sufficient** duration to produce the needed delay, so that actual delay units may not be necessary.
 - Thus, an asynchronous sequential circuit may be regarded as a **combinational circuit with feedback.**

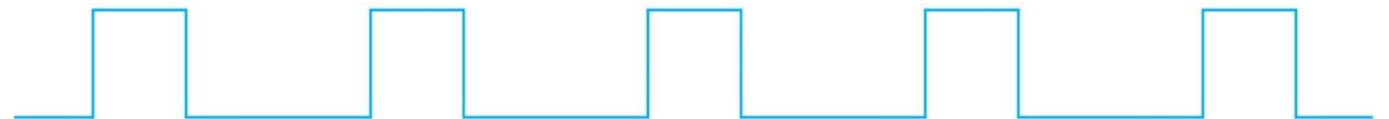
Synchronous sequential circuits

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- The storage elements (memory) used in clocked sequential circuits are called **flipflops**.
- A flip-flop is a binary storage device capable of storing **one bit of information**
- **In a stable state**, the output of a flip-flop is either **0** or **1**
- The **new value** is stored (i.e., the flip-flop is updated) when a pulse of the clock signal occurs



(a) Block diagram

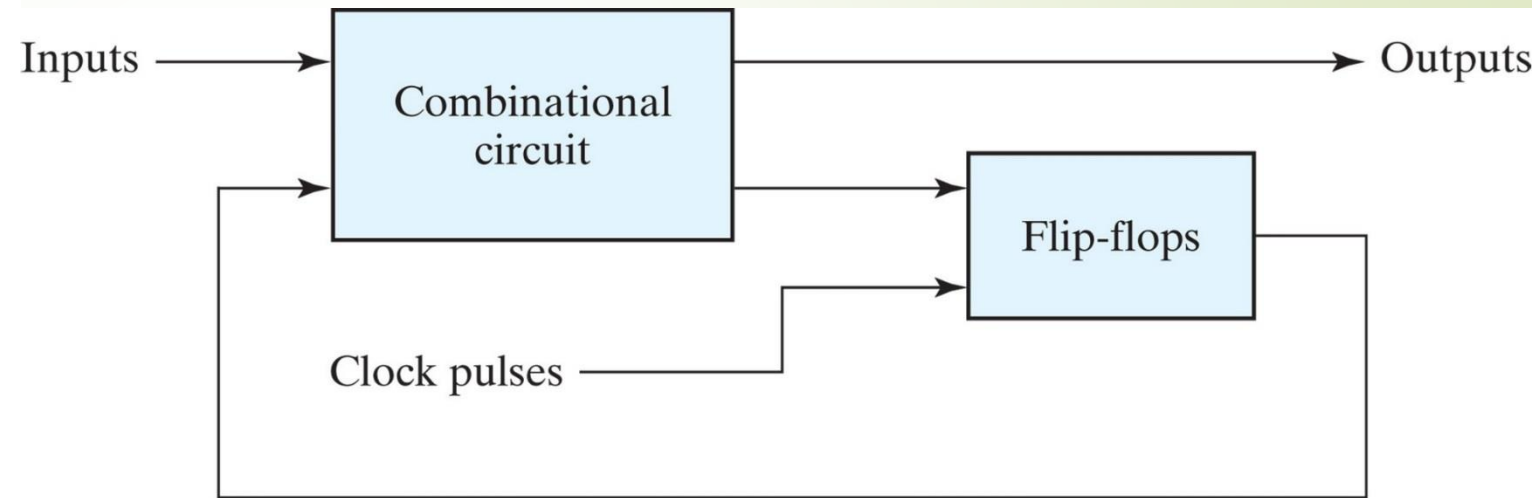


(b) Timing diagram of clock pulses

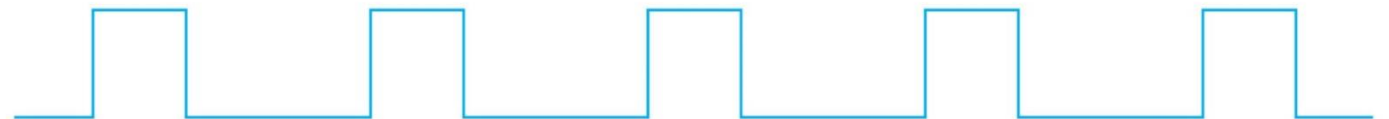
Synchronous sequential circuits

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- Prior to the occurrence of the clock pulse, the combinational logic forming the next value of the flip-flop must have reached a **stable value**.
- Consequently, the **speed** at which the **combinational logic circuits** operate is critical
- the combinational logic must respond to a change in the state of the flip-flop in time to be updated before the next pulse arrives.



(a) Block diagram



(b) Timing diagram of clock pulses

5.3 Storage Elements: Latches

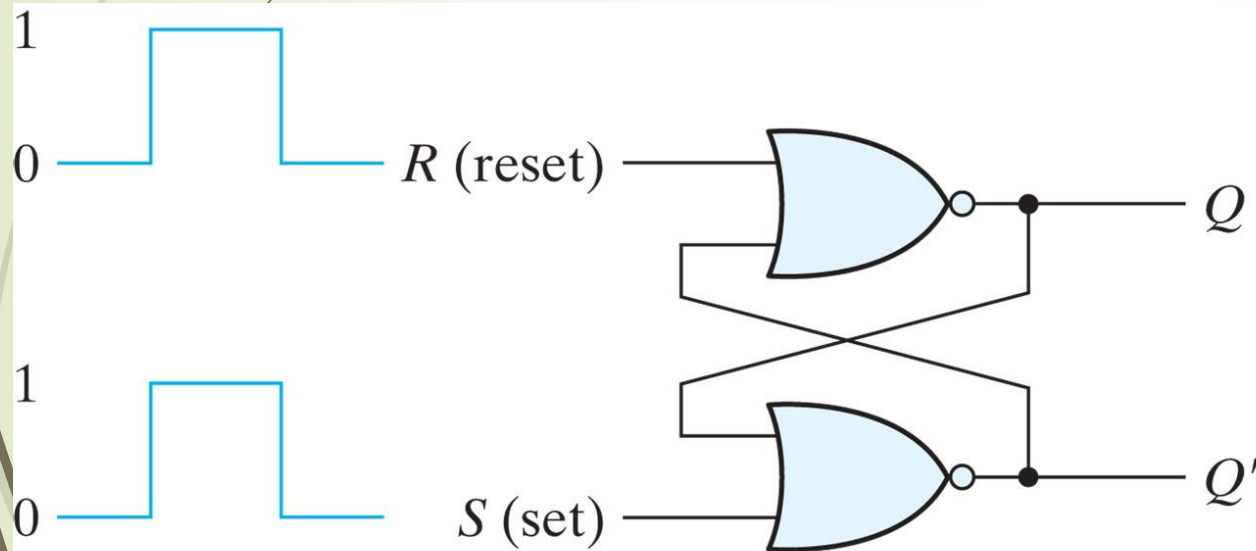
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- *Latches: Storage elements that **operate with signal levels** (rather than signal transitions) are referred to as **latches** ;*
 - Latches are said to be **level sensitive devices**.
 - Useful in asynchronous sequential circuits
 - Not practical for use as storage elements in synchronous sequential circuits
- *Flip-flops: Those controlled by a **clock** transition are **flip-flops**.*
 - flip-flops are **edge-sensitive devices**.

SR Latch (Active High)

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- The SR latch is a circuit with **two cross-coupled NOR gates** or **two cross-coupled NAND gates**, and **two inputs** labeled **S** for **set** and **R** for **reset**.



(a) Logic diagram

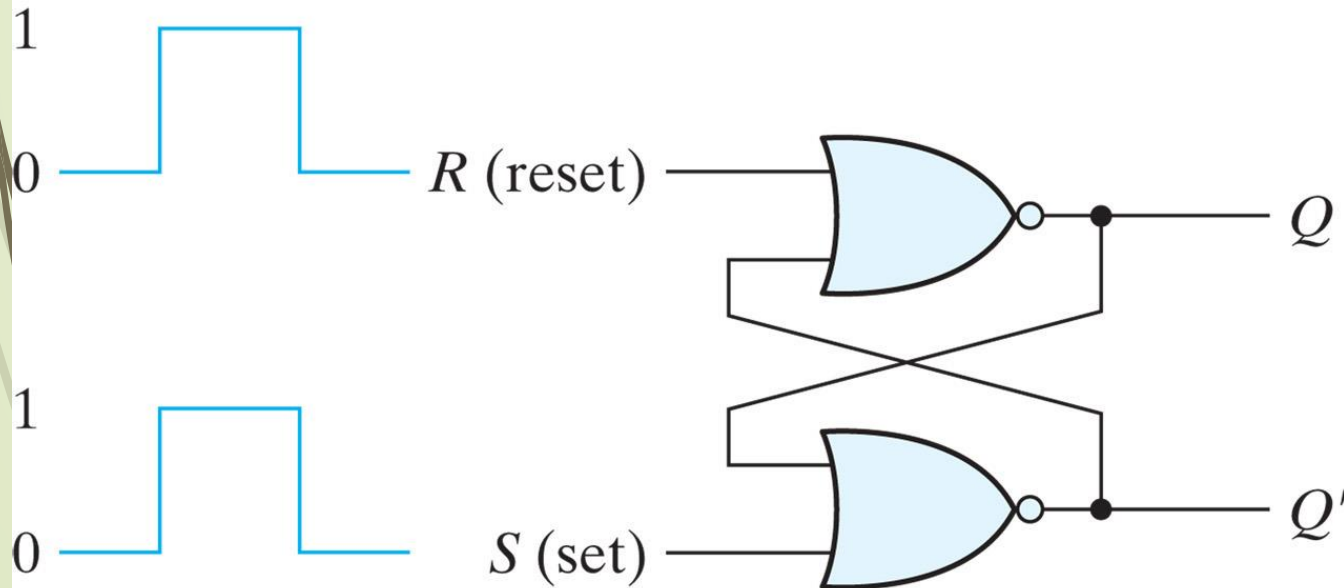
S	R	Q	Q'	
1	0	1	0	
0	0	1	0	(after $S = 1, R = 0$)
0	1	0	1	
0	0	0	1	(after $S = 0, R = 1$)
1	1	0	0	(forbidden)

(b) Function table

SR Latch with NOR gate

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- The latch has two useful states.
- When output $Q = 1$ and $Q' = 0$, the latch is said to be in the **set state**
- When $Q = 0$ and $Q' = 1$, it is in the **reset state**
- Under normal conditions, both inputs of the latch remain at 0 unless the state has to be changed.
- The application of a momentary **1** to the **S** input causes the latch to go to the set state
- **Removing the active input from S leaves the circuit in the same state**

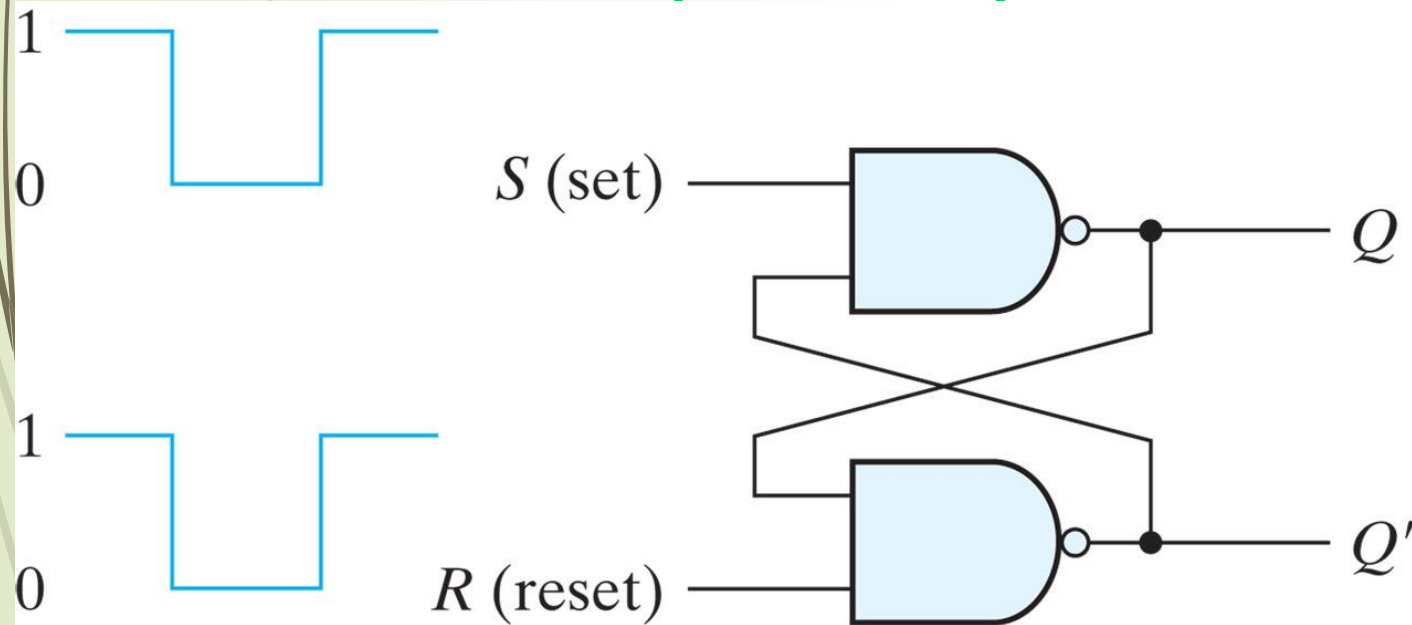


S	R	Q	Q'	
1	0	1	0	
0	0	1	0	(after $S = 1, R = 0$)
0	1	0	1	
0	0	0	1	(after $S = 0, R = 1$)
1	1	0	0	(forbidden)

SR Latch with NAND gate (Active Low)

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- Because the NAND latch requires a 0 signal to change its state, it is sometimes referred to as an $S'R'$ latch
- **Video-1** <https://www.youtube.com/watch?v=-aQH0ybMd3U>



(a) Logic diagram

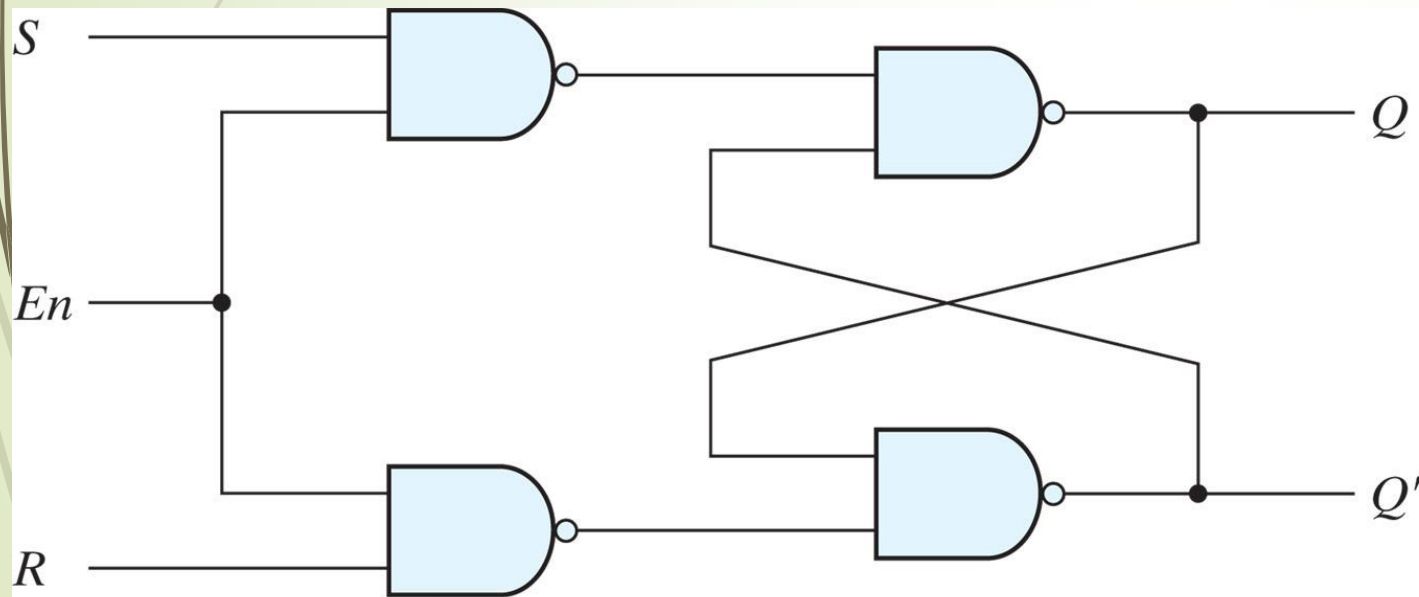
S	R	Q	Q'	
1	0	0	1	
1	1	0	1	(after $S = 1, R = 0$)
0	1	1	0	
1	1	1	0	(after $S = 0, R = 1$)
0	0	1	1	(forbidden)

(b) Function table

SR latch with control input (Gated)

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- The outputs of the NAND gates stay at the logic-1 level as long as the enable signal remains at 0.
- Video-2
https://www.youtube.com/watch?v=HxAhOETcyr4&ab_channel=ComputerScience



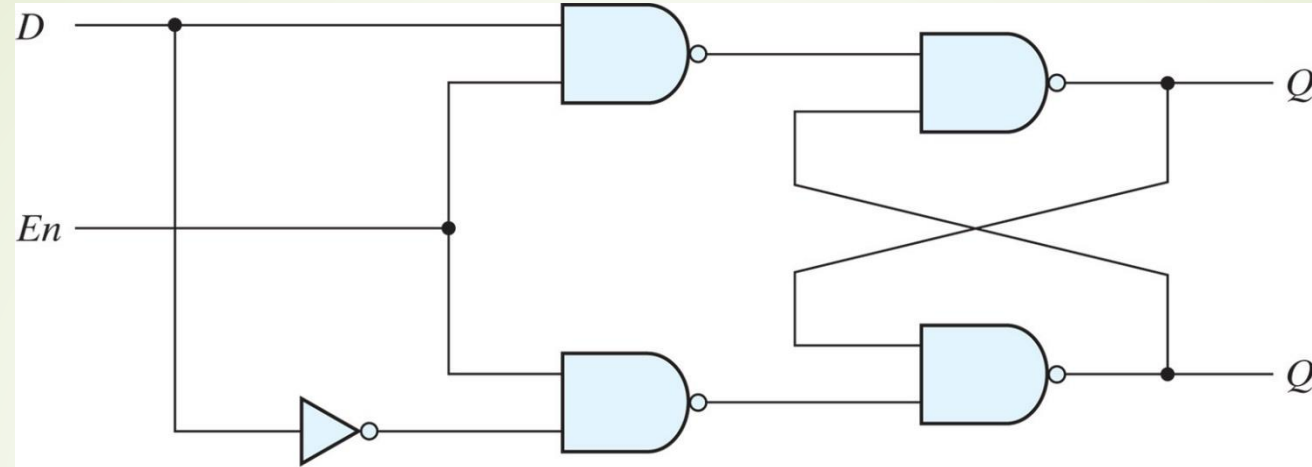
(a) Logic diagram

En	S	R	Next state of Q
0	X	X	No change
1	0	0	No change
1	0	1	$Q = 0$; reset state
1	1	0	$Q = 1$; set state
1	1	1	Indeterminate

(b) Function table

D Latch (Transparent Latch)

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(a) Logic diagram

En	D	Next state of Q
0	X	No change
1	0	$Q = 0$; reset state
1	1	$Q = 1$; set state

(b) Function table

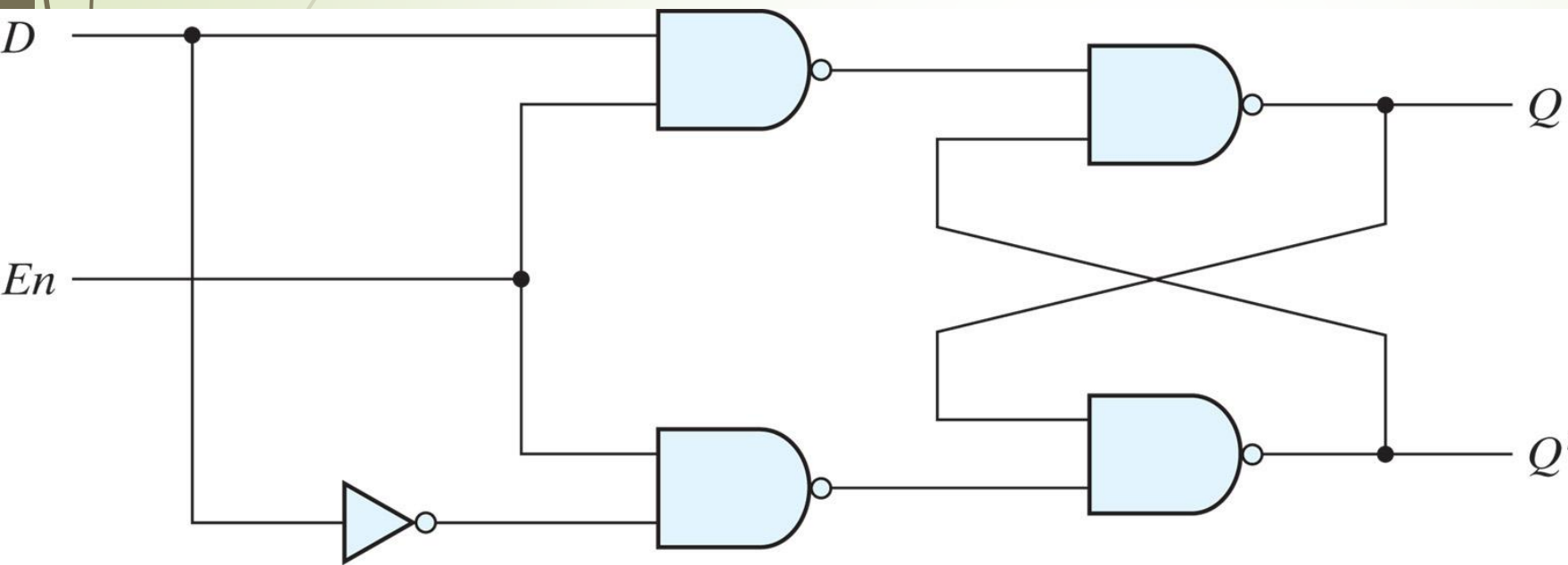
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- It is suited for use as a temporary storage for **binary information**
- The **binary information** present at the data input of the D latch is **transferred to the Q output** when the **enable input is asserted**
- The **output follows changes** in the data input as long as the enable input is asserted
- For this reason, the circuit is often called a **transparent latch**
- When the enable input signal is de-asserted, the **binary information** that was present at the data input at the time the transition occurred **is retained**

D Latch (Transparent Latch)

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- One way to eliminate the undesirable condition of the indeterminate state in the SR latch is to ensure that inputs S and R are never equal to 1 **at the same time**
- Video-3 https://www.youtube.com/watch?v=y7Zf7Bv_J74



(a) Logic diagram

En	D	Next state of Q
0	X	No change
1	0	$Q = 0$; reset state
1	1	$Q = 1$; set state

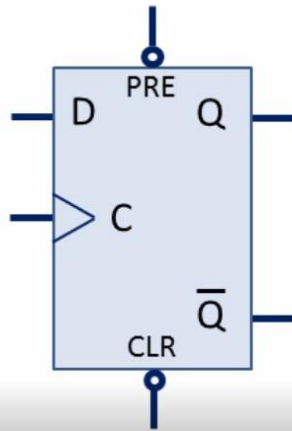
(b) Function table

Clocked D Latch

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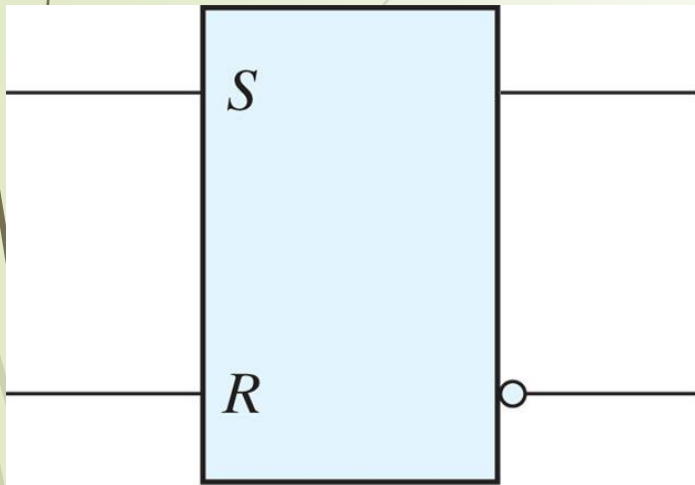
- Controlling over the behavior of the latch
- **Video 4 - <https://www.youtube.com/watch?v=8bUKw2cGcGg>**

Clocked D Latch

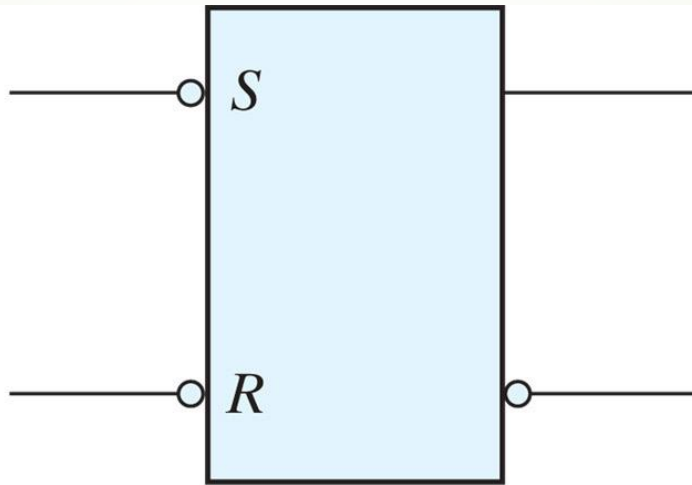


Different Latch Symbols

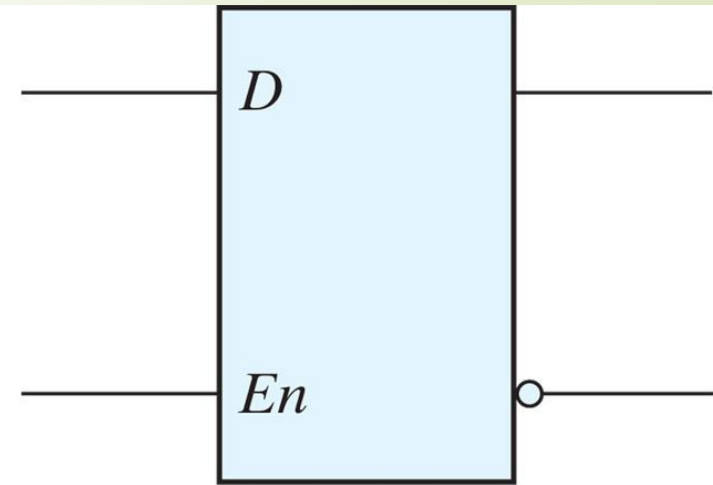
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SR



$\overline{S}\overline{R}$



D